

Building Detection from Satellite Imagery using YOLOv8

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1 Problem

The goal of this project is to detect buildings from satellite images using a deep learning model. Given a satellite image, the model identifies buildings and draws bounding boxes around them. The model can also estimate the number of buildings present in an image.

Building detection is useful in applications such as urban planning, mapping, infrastructure monitoring, and disaster assessment.

2 Dataset

The SpaceNet Buildings dataset from Kaggle was used for this project. The dataset contains satellite images along with YOLO-format annotation files.

The dataset was divided into:

- Training Images: 918
- Validation Images: 115
- Test Images: 115

A random sample of training images was visualized with their bounding boxes. The dataset contains images with varying numbers of buildings, ranging from sparse regions to dense urban areas.

Some challenges observed in the dataset include:

- Small building rooftops
- Dense clusters of buildings
- Shadows covering buildings
- Buildings near image boundaries

3 Model

YOLOv8s was selected for building detection. YOLO (You Only Look Once) is a popular object detection model because it provides a good balance between speed and accuracy.

Pretrained weights (`yolov8s.pt`) were used instead of training from scratch. This allows the model to learn faster and achieve better performance.

The model was trained to detect one class:

- Building

4 Training

Training was performed on Google Colab using a Tesla T4 GPU.

The following settings were used:

- Epochs: 20
- Batch Size: 16
- Image Size: 640×640

The model completed training successfully.

Final training metrics:

- Final Box Loss: 1.25079
- Final mAP@50: 0.82721

4.1 Training Curves

The validation loss followed the training loss closely throughout training, suggesting stable learning. The mAP@50 increased steadily and eventually stabilized, indicating convergence of the model.

5 Results

The model achieved the following performance on the validation set:

Metric	Value
Precision	0.8313
Recall	0.7498
mAP@50	0.8276
F1 Score	0.7885

Table 1: Model Performance Metrics

These results show that the model was able to detect buildings accurately while maintaining a good balance between precision and recall.

6 Observations

Some observations from the results are:

- The model performs well on clearly visible buildings.
- Buildings in dense urban regions are generally detected successfully.
- Very small buildings are occasionally missed.
- Shadows and overlapping structures sometimes reduce detection accuracy.

Possible improvements include:

- Training for more epochs.
- Using a larger YOLOv8 model.
- Applying additional data augmentation techniques.

7 Conclusion

In this project, a YOLOv8-based building detection system was developed using satellite imagery.

The model achieved an mAP@50 of 0.8276, a precision of 0.8313, and a recall of 0.7498. The results demonstrate that YOLOv8 can effectively detect and count buildings from satellite images.

Overall, the model showed good performance and can serve as a foundation for more advanced remote sensing applications.